

Structured summary: Three-component decomposition of treatment response in aggregated N-of-1 trials: pharmacological, expectation-driven, and natural-history components

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34 *This document is a structured, section-by-section summary of the key points argued in the*
 35 *companion manuscript, “Three-component decomposition of treatment response in aggregated*
 36 *N-of-1 trials: pharmacological, expectation-driven, and natural-history components.” It is pro-*
 37 *vided as a supplementary reading aid and does not stand on its own; all notation, derivations,*
 38 *simulation detail, and citations are in the main manuscript.*

39

40 **0.1 A claim about how N-of-1 trials should be analyzed**

41 **This paper argues that treatment response in N-of-1 trials comprises three**
 42 **causally distinct components that must be analyzed jointly rather than as a**
 43 **single indicator.**

- 44 • The three components are *BR* (pharmacological), *PB* (placebo-belief), and *TV* (natural
45 history).
- 46 • Each component is causally distinct, clinically interpretable, and identifiable from a
47 well-designed N-of-1 trial.
- 48 • The standard single-indicator mixed-effects model conflates all three and renders
49 biomarker validation unanswerable.

50 **The contribution is the consolidation and extension of an established structural-**
 51 **decomposition tradition, not a novel decomposition.**

- 52 • The additive disease-progression, placebo, and drug-action decomposition is well estab-
53 lished in pharmacometrics and longitudinal trial methodology.
- 54 • The present framework extends that tradition with an explicit, separately identified
55 placebo-belief channel and brings it to the aggregated N-of-1 setting.
- 56 • The decomposition logic is not specific to N-of-1; it applies wherever a longitudinal
57 treatment response is analyzed.

58 **0.2 Why this matters: three failure modes of the one-component analysis**

59 **The one-component analysis has three documented failure modes that grow with**
 60 **unmodeled component heterogeneity.**

- 61 • The three failure modes are placebo-attribution bias, regression-to-the-mean, and
62 biomarker-validation failure.
- 63 • Each failure mode is documented in the chronic-condition trial literature, not merely
64 hypothetical.
- 65 • The failures are present regardless of sample size.

66 **Placebo-attribution bias is sample-size-invariant and documented across antide-**
67 **pressant, anxiety, and IBS literatures.**

- 68 • Lumped treatment-effect estimates overstate pharmacological efficacy in proportion to
69 PB magnitude and variance.
- 70 • Kirsch et al. and Locher et al. demonstrated that antidepressant apparent efficacy is
71 largely explained by placebo response and natural fluctuation.
- 72 • Placebo responsiveness has measurable biological correlates (e.g. COMT genotype), con-
73 firming it is a structured phenomenon, not noise.

74 **Regression-to-the-mean contributes several points of apparent improvement in**
75 **early trial weeks and is absorbed without trace by the one-component analysis.**

- 76 • Enrolment at peak severity guarantees a natural-history regression component irrespec-
77 tive of pharmacology.
- 78 • Hrobjartsson et al. systematic reviews across 200+ trials showed much of conventional
79 placebo response was indistinguishable from natural-history drift.
- 80 • The one-component analysis has no formal mechanism to recover the pharmacological
81 effect once regression is absorbed.

82 **Biomarker validation is invalidated when a biomarker correlates with PB or TV ,**
83 **rendering precision-medicine stratification wrong regardless of trial size.**

- 84 • A biomarker correlated with placebo responsiveness or natural-history trajectory is in-
85 distinguishable from a true pharmacological predictor in a one-component analysis.
- 86 • The PATH statement defers mechanistic attribution to domain knowledge; the BR-PB-
87 TV decomposition supplies that substrate.
- 88 • Where such coupling cannot be ruled out, rigorous pharmacological-biomarker validation
89 requires the decomposition; where the biomarker is known orthogonal to PB and TV ,
90 the one-component analysis already recovers the pharmacological slope.

91 **0.3 The current state of the N-of-1 literature**

92 **The N-of-1 literature has matured substantially but motivates three observations**
93 **that the present paper addresses.**

- 94 • The design has evolved from individual-patient consultation to a framework supporting
95 population-level inference.
- 96 • CONSORT reporting standards and quality audits document progressive improvement
97 in analytic transparency.
- 98 • Three specific gaps in the literature motivate the decomposition framework proposed
99 here.

100 **No published N-of-1 trial employs a structural decomposition of treatment re-**
101 **sponse despite three decades of methodological literature.**

- Audit data show the field has relied on paired t-tests, binary-indicator linear mixed-effects models (LME), and hierarchical Bayesian regressions.
- Recent methodological advances (Liao 2023, Blackston 2019, Carrozzo 2025) retain the one-component response framing.
- The decomposition gap persists even as carryover and identifiability problems are increasingly recognized.

Placebo-component separation is widely recognized but almost entirely unaddressed at the protocol level in N-of-1 methodology.

- The Hendrickson hybrid blinded-discontinuation phase is among the few protocol mechanisms that supply *PB*-versus-*BR* identifiability.
- Placebo run-in, open-label-placebo, and mediator-analysis approaches have not been imported into N-of-1 designs.
- The gap between awareness of the problem and protocol-level solutions motivates the present framework.

Kaptchuk’s within-placebo decomposition is the closest structural analogue; no published paper applies the corresponding full-response decomposition to N-of-1 trials.

- Kaptchuk et al. (2008) empirically decomposed the placebo response into observation, ritual, and relationship components.
- Open-label placebo trials confirm that clinically meaningful PB-only response is possible even when patients know they receive placebo.
- The BR-PB-TV framework is structurally analogous but operates at the level of the full treatment response.

The decomposition gap is real, the tools to fill it are mature, and the failure modes are large enough to make it a methodological priority.

- The gap between available methodology and current practice is documented and not attributable to missing tools.
- Failure modes in chronic-condition pharmacology are consequential, not academic.
- The remainder of the paper develops, applies, and evaluates the decomposition framework.

0.4 The neurobiological and epidemiological reality of the components

Each component of the decomposition reflects a real causal mechanism documented in independent literatures.

- *BR* corresponds to pharmacological action; *PB* to neurobiologically real belief-driven mechanisms; *TV* to natural-history drift.
- The components are not statistical constructs but distinct causal pathways with separate literatures.

139 • The section reviews the evidence for each component before establishing the formal
140 model.

141 ***BR* is the pharmacological target of regulation; *PB* is neurobiologically real and**
142 **phase-modulated, which is precisely the design contrast that makes the decom-**
143 **position identifiable.**

- 144 • *PB* is mediated by endogenous opioid/dopamine release, HPA-axis regulation, and top-
145 down attentional shifts.
- 146 • Wager et al. neurologic pain signature provides a clean fMRI dissociation between phar-
147 macological and expectancy contributions.
- 148 • Phase-dependent belief variation (open-label vs. blinded) is the design contrast that
149 supplies *PB* identifiability.

150 ***TV* is the untreated counterfactual trajectory; its heterogeneity is the central**
151 **object of precision-medicine inference and must be modeled, not absorbed into**
152 **noise.**

- 153 • *TV* sign depends on disease class: positive for acute recovery, near-zero for stable chronic,
154 negative for progressive conditions.
- 155 • Selection-driven enrollment at peak severity guarantees positive *TV* in early trial weeks
156 regardless of disease class.
- 157 • The PATH statement identifies heterogeneity of treatment effects (HTE) modeling as
158 mandatory for precision-medicine inference.

159 0.5 What this paper contributes

160 **The paper makes four contributions spanning formalization, identifiability map-**
161 **ping, tractable methodology, and quantification of the inferential cost of the**
162 **one-component analysis.**

- 163 • Sections 2–3 formalize the decomposition; section 4 maps it to design contrasts; section
164 5 develops tractable analysis methods.
- 165 • Sections 6–7 provide worked examples and a Monte Carlo simulation study; section 8
166 applies the framework to biomarker validation.
- 167 • A companion pedagogical document provides extended worked examples for first-time
168 readers.

169 1 Notation and the formal model

170 **This section fixes compact notation and states the formal model; a fuller worked-**
171 **example treatment is in the companion pedagogical document.**

- 172 • Definitions are precise enough to support the identifiability and analysis-side arguments
173 in later sections.

- The presentation is kept compact; extended derivations are deferred to the companion document.
- The companion pedagogical document provides numerical worked examples at each formal step.

where Y_{it} is the observed symptom score, BL_i is participant i 's baseline score, the three components BR , PB , TV are non-negative reductions in symptom severity attributable to the three causal mechanisms identified in section 1, and ε_{it} is observation-level noise. The sign convention is that an increase in any component reduces Y ; all three components are zero at $t = 0$ by construction (no exposure yet to drug, no belief activation, no elapsed natural history) and grow over the duration of the trial.

Each component is driven by its own time-on-state variable, making the components structurally distinct even when they overlap in calendar time.

- BR_{it} accumulates with time-on-drug; PB_{it} with time-on-belief scaled by expectancy $\eta(\text{phase})$; TV_{it} with calendar time from trial entry.
- The distinct time indices are the structural source of identifiability from phase-contrast data.
- All components are zero at $t = 0$ by construction and grow over the trial duration.

rescaled so that $C(0) = 0$ and $C(t) \rightarrow m_C(1 - e^{-d_C t})$ as $t \rightarrow \infty$, for $C \in \{BR, PB, TV\}$. The three Gompertz parameters (m_C, d_C, r_C) are participant-specific random effects drawn from population distributions, and the participant-level heterogeneity in these distributions is the principal source of between-participant response variability that the framework is designed to characterize.

The residual term has AR(1) structure and the three components are permitted to co-vary through cross-component covariances.

- Within-participant residuals follow $\text{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2 \rho^{|h|}$.
- Within-time and across-time cross-component covariances permit participant-level heterogeneity in one component to co-vary with the others.
- This structure accommodates biologically plausible correlations, such as high- BR patients also tending toward high- PB .

2 The three components

This section presents each component with its biological motivation, Gompertz parameterization, and identifiability conditions to establish the model as causally interpretable.

- The three components are presented in turn: BR (biological response), PB (placebo-belief), TV (natural history).
- Each component is grounded in independent biological literature before its parametric form is stated.

- The section’s purpose is to demonstrate that the model sum reflects causal structure, not mathematical convenience.

2.1 Biological response (BR)

***BR* is the pharmacological target of regulatory approval and biomarker stratification, independent of belief or natural history.**

- *BR* arises from chemical action on the body and is zero for inert pills regardless of patient expectation.
- It is the estimand that approval decisions, dose selection, and biomarker-guided stratification require.
- Separating *BR* from *PB* and *TV* is the central precision-medicine problem the framework addresses.

Pharmacokinetic/pharmacodynamic (PK/PD) considerations produce a saturating *BR* trajectory with a slow initial rise, a peak-growth phase, and a sharp drop to zero on discontinuation.

- Plasma steady state is reached within a few half-lives; pharmacodynamic integration (e.g. receptor adaptation) adds additional lag.
- For prazosin-PTSD, Raskind et al. (2013) calibrate the onset to several weeks of sustained exposure.
- The saturating shape motivates the Gompertz over linear or simple exponential alternatives.

The three Gompertz parameters (m_{BR}, r_{BR}, d_{BR}) capture ceiling, onset rate, and climb shape respectively, calibrated to prazosin-PTSD reference values.

- m_{BR} is the biological ceiling (maximum reduction at infinite exposure); r_{BR} the onset rate; d_{BR} the displacement shaping the early climb.
- Prazosin-PTSD calibration yields $m_{BR} \approx 11$, $r_{BR} \approx 0.42/\text{week}$, $d_{BR} \approx 5$.
- These are participant-specific random effects whose population distributions encode between-participant response heterogeneity.

The Gompertz is preferred over linear and simple-exponential alternatives because it captures the asymmetric slow-start saturation profile of chronic-condition pharmacodynamics.

- Linear $BR = \beta t$ is biologically implausible past the first few weeks due to unbounded growth.
- Exponential $BR = m(1 - e^{-rt})$ lacks the slow-start phase from receptor adaptation and gene-expression changes.
- The Gompertz interpolates between both limits and its three-parameter form avoids overfitting; sensitivity analysis is reported separately in 07-gompertz-evaluation.

247 *BR* is identified by the blinded-discontinuation contrast, where drug removal col-
248 lapses *BR* to zero while *PB* remains intact.

- 249 • Within-participant on-drug versus off-drug contrasts are the primary identification mech-
250 anism.
- 251 • The blinded-discontinuation phase provides the cleanest contrast: *BR* drops, *PB* is
252 preserved.
- 253 • This contrast does not require the analyst to model *PB* or *TV* explicitly to recover *BR*.

254 2.2 Placebo-belief response (PB)

255 *PB* is the strict placebo response: improvement that persists when the active
256 drug is silently replaced, provided the patient's belief is unchanged.

- 257 • *PB* is belief-dependent, not drug-dependent; it would survive a silent switch to placebo
258 under sustained belief.
- 259 • It is distinct from *BR* in causal origin: *BR* requires chemical action; *PB* requires only
260 belief.
- 261 • The definition is strict and operational, not vague or colloquial.

262 *PB* reflects neurobiologically real mechanisms triggered by belief, not pharma-
263 cology; neuroimaging and genetic evidence confirm it is a structured biological
264 phenomenon.

- 265 • Belief-driven endogenous opioid/dopamine release, HPA-axis modulation, and top-down
266 attentional shifts are all documented mechanisms.
- 267 • Wager et al. neurologic pain signature dissociates pharmacological from expectancy con-
268 tributions via fMRI.
- 269 • COMT genotype predicts placebo responsiveness, confirming biological structure rather
270 than measurement artifact.

271 *PB* is identifiable from *BR* only when belief and treatment allocation differ, as in
272 the blinded-discontinuation phase.

- 273 • The blinded-discontinuation phase is the canonical contrast: patients cannot tell whether
274 they remain on active drug.
- 275 • Belief drops to a partial level ($\eta \approx 0.5$) because the patient's expectation is hedged, not
276 absent.
- 277 • Designs without a phase that decouples belief from treatment cannot separately identify
278 *PB* from *BR*.

279 The $\eta = 0.5$ blinded-phase expectancy is a modeling assumption operationalizing
280 response expectancy, not a measured quantity, and requires sensitivity analysis.

- 281 • The construct is response expectancy in the sense of Kirsch (1985); to our knowledge
282 the residual expectancy under blinded discontinuation has not been measured directly.

- Open-label placebo trials confirm clinically meaningful *PB*-only responses even under explicit disclosure, bounding the residual response away from zero.
- Sensitivity analyses varying η are mandatory before clinical conclusions depend on this single parameter.

Kaptchuk et al. (2008) provide the structural precedent; the BR-PB-TV framework extends the same decomposition logic to the full treatment response.

- Kaptchuk decomposed the placebo response itself into observation, ritual, and relationship components empirically.
- The present framework applies analogous decomposition logic at the level of drug, placebo, and natural history.
- Both frameworks reject the 'expectancy lump' model in favor of identifiable additive parts.

***PB* variance scales with η , not just its mean; ignoring this heteroscedastic structure miscalibrates standard errors and downstream biomarker tests.**

- Phase-dependent variance follows $SD(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$.
- Open-label phases have high between-participant *PB* variance; blinded phases have low variance.
- Ignoring heteroscedasticity deflates standard errors in open-label and inflates them in blinded phases, with consequences for CI coverage and biomarker test calibration.

2.2.1 Additivity, subadditivity, and the BR-PB interaction

Additivity is a modeling assumption, not biological fact; the section examines whether a $BR \times PB$ interaction term is warranted.

- The formal model writes total response as a sum; this section asks whether the *BR-PB* interaction is zero.
- The question has inferential consequences for biomarker validation and for what the phase-augmented analysis captures.
- The treatment is preparatory for the generalized interaction model introduced later in this section.

The additive model may be wrong: published literature documents subadditivity between pharmacological and placebo-belief effects in some clinical contexts.

- Subadditivity means the combined $BR + PB$ effect is less than the sum of each component measured separately.
- Evidence from Kaptchuk et al. and Benedetti et al. shows the phenomenon is context-dependent, not universal.
- Formal test: $\partial Y / \partial (BR \cdot PB) > 0$ in a saturated mean-structure model.

Three candidate mechanisms for subadditivity are shared neural substrate, scale ceiling effects, and belief-dynamics adaptation.

- Shared pain-modulation circuits produce saturating occupancy when both BR and PB activate them simultaneously.
- Scale ceiling effects prevent registration of additional symptom reduction once $BR + PB$ saturates the measurement range.
- Belief-dynamics adaptation: unexpectedly large BR can recalibrate patient expectation downward, attenuating ongoing PB .

where $\gamma < 0$ encodes subadditivity, $\gamma = 0$ recovers the additive form, and $\gamma > 0$ encodes superadditivity, is the natural extension. The interaction coefficient γ is identifiable from the same trial-design contrasts that identify the marginal components: open-label phases supply $BR \cdot PB$ products at high η , blinded phases supply $BR \cdot PB$ products at attenuated η , and the γ coefficient is estimated from the difference between the two regimes once the marginal components are accounted for.

Subadditivity biases biomarker predictive accuracy estimates and is partly absorbed by the phase-augmented analysis even without explicit γ estimation.

- Under subadditivity, the biomarker-by- BR association is partly cancelled, inflating apparent predictive accuracy in the additive model.
- Phase designs attenuating PB (blinded, crossover) push the effective biomarker-treatment estimate closer to the true biomarker-by- BR value.
- The phase-augmented analysis in section 5 absorbs subadditivity as a side benefit.

A forthcoming simulation study will quantify how bias in the biomarker-treatment interaction scales with γ ; the pre-registration predicts linear scaling.

- Bias in $\hat{\beta}_{bm:D}$ is expected to scale linearly with γ at small $|\gamma|$.
- Under subadditivity, bias sign is opposite to the true biomarker-by- BR effect; under superadditivity it is aligned.
- Empirical quantification is deferred to the production simulation program in section 7.

Non-zero $\hat{\gamma}$ does not establish true subadditive interaction; latent-class structure with class-correlated components produces the same marginal pattern.

- If class membership correlates with both BR and PB strength, the marginal covariance of $BR \cdot PB$ mimics a direct interaction term.
- This confounding is invisible from the one-component analysis perspective but is identifiable in a class-aware analysis.
- Apparent subadditivity can arise from latent population heterogeneity with no direct BR - PB interaction in the class-conditional data-generating process (DGP).

Distinguishing true subadditivity from latent-class confounding requires either a class-aware analysis or the phase contrasts the BR-PB-TV framework provides.

- The two mechanisms produce qualitatively similar marginal effects but are inferentially distinct.

- $\hat{\gamma}$ should be reported alongside a class-aware sensitivity analysis, not as standalone evidence for mechanistic subadditivity.
- The latent-class paper's `pb_class_correlation` simulation axis is the direct test; cross-paper synthesis is the cleanest disentanglement.

2.3 Time-variant natural-history component (TV)

***TV* is the untreated counterfactual trajectory; it is participant-specific, structured, and can be positive, negative, or near-zero depending on disease class.**

- *TV* is the symptom change that would have occurred without treatment, integrated over the trial duration.
- It varies in sign across disease classes and across patients within a class.
- It is independent of treatment status and belief, distinguishing it structurally from *PB*.

***TV* is a structured participant-specific signal, not noise; its distinguishing feature from *PB* is independence from treatment status and belief.**

- Unlike ε_{it} , *TV* is participant-level (slow trajectory) and requires explicit modeling, not absorption into noise.
- Off-drug baselines and follow-up windows are the primary data sources for estimating *TV*.
- *TV* is modulated by neither treatment assignment nor patient belief, whereas *PB* is modulated by both.

2.3.1 When does natural history actually exist?

The shape of *TV* depends on disease class, trial timescale, and selection process; none of these can be assumed and all must be recovered empirically.

- The assumption that subjects improve over time is true for some conditions and systematically wrong for others.
- Disease class, trial timescale, and enrollment selection jointly determine the sign and magnitude of *TV*.
- Recovering the empirical shape of *TV* is one of the decomposition's primary objectives.

***TV* sign varies systematically by disease class: positive for acute recovery, near-zero for stable chronic, negative for progressive, and structured for cyclical conditions.**

- Acute conditions (post-operative pain, acute back pain) resolve intrinsically; substantial *TV*-improvement appears in the placebo arm.
- Progressive conditions (Alzheimer's, ALS) require explicit negative-trajectory modeling; apparent treatment effect is slowing of progression.
- Cyclical/triggered conditions (PTSD with trauma anniversaries, relapsing-remitting MS) require covariates beyond the Gompertz form.

393 **Selection-induced TV is real and rarely zero even for stable chronic conditions,**
394 **because enrollment coincides with peak severity triggering help-seeking.**

- 395 • Regression to the mean alone contributes several points of apparent improvement in
396 early trial weeks.
- 397 • Hrobjartsson et al. systematic reviews across 200+ trials show much of conventional
398 placebo response is indistinguishable from natural-history drift.
- 399 • Kirsch (2008) and Locher (2015) demonstrate the clinical consequences in antidepressant
400 and late-life depression trials.

401 **Trial participation itself induces TV -improvement through Hawthorne,**
402 **structured-visit, and diary-completion effects, independent of pharmacology**
403 **or placebo.**

- 404 • Hawthorne, structured-visit, and diary-completion effects each contribute symptom im-
405 provements attributable to participation, not treatment.
- 406 • These contributions are absorbed into TV in the decomposition, not into BR or PB .
- 407 • Ignoring these participation-induced effects inflates the apparent pharmacological and
408 placebo components.

409 **2.3.2 Can we assume no TV effect?**

410 **Assuming $TV = 0$ is rarely safe in chronic psychiatric or pain trials; the conser-**
411 **vative default is to estimate TV from the data.**

- 412 • Zero- TV assumption is valid only when all four conditions hold: stable disease, short
413 trial, non-severity-peaked recruitment, no ancillary care.
- 414 • For most chronic conditions at least one condition fails; the assumption should be tested,
415 not asserted.
- 416 • Statistical efficiency lost by estimating TV is smaller than the bias introduced by as-
417 suming it zero.

418 **The decomposition makes the zero- TV assumption testable and shows that ab-**
419 **sorbing TV into residuals produces autocorrelated, heteroscedastic residuals that**
420 **bias BR and PB standard errors.**

- 421 • $\hat{m}_{TV}$ with a variance estimate from the model provides an empirical test of the zero- TV
422 assumption.
- 423 • Participant-level TV is structured and correlated with the on/off-drug schedule; it is
424 not independent noise.
- 425 • Lumping TV into ε produces autocorrelated within-participant residuals and het-
426 eroscedastic cross-participant residuals.

427 **2.3.3 Why a Gompertz, not a linear time covariate?**

428 **A linear week covariate is insufficient for TV because it enforces linearity, ignores**
429 **participant heterogeneity, and cannot distinguish TV from BR without the explicit**

decomposition.

- A **week** covariate constrains each additional week to contribute a fixed natural-history change, misrepresenting curved or saturating disease trajectories.
- A single coefficient estimates a population-mean trajectory, ignoring participant-level heterogeneity in natural history.
- Without the decomposition and design contrasts, **week** absorbs both *BR* and *TV*, making them inseparable.

3 Identifiability and trial-design contrasts

Identifiability is supplied by trial design; each phase of the hybrid design contributes a different combination of components to the observed trajectory.

- The decomposition is mathematically specified but requires trial-design contrasts to be statistically identified.
- This section characterizes the hybrid open-label-plus-blinded-discontinuation design as the primary identifiability vehicle.
- Each of the three phases supplies a distinct mixture of *BR*, *PB*, and *TV*.

Three phase-by-allocation contrasts isolate *PB*, *BR*, and *TV* respectively; the LME model combines them but the inference is driven by design.

- Open-label versus blinded contrast yields approximately $0.5 \cdot PB$, isolating the belief component conditional on a known blinded-phase η .
- Blinded on-drug versus blinded placebo yields *BR* once drug carryover has elapsed (since *PB* and *TV* match across arms within the blinded phase).
- Off-drug phases yield *TV* plus residual off-drug *PB*, with no active *BR* contribution beyond carryover.

Designs lacking specific phases lose identifiability of the corresponding component; the decomposition machinery is only as useful as the design allows.

- A pure open-label design cannot separate *BR* from *PB* because both are present at full strength throughout.
- A pure blinded design cannot estimate $\eta = 1$ *PB* because no phase achieves full-strength belief.
- Trial designers should verify each component has at least one phase in which it is uniquely present before committing to the design.

Design-contrast identifiability is necessary but not sufficient: the three Gompertz trajectories are collinear, so joint estimation is credible only under specific design and sample-size conditions.

- The components share an onset shape and overlap in calendar time, so their parameters can be near-collinear even when the identifying contrasts are present.

- Joint recovery requires a phase in which each component varies while the others are held fixed, and an off-drug window long enough for TV to separate from a decaying BR .
- The recoverable region of the design-and-sample-size space is mapped by the pre-registered identifiability diagnostics, not asserted.

4 Analysis-side methodology

Four constraints make a component-matched analysis impractical at trial-relevant sample sizes; a phase-augmented LME captures the important part of the gap at minimal cost.

- The question of whether to mirror the DGP in the analysis model is reasonable; the section addresses it directly.
- The four constraints are identifiability, degrees of freedom, convergence fragility, and marginal estimand robustness.
- A phase-by-treatment indicator is the recommended modest extension that captures BR-versus-PB sensitivity without the full parametric cost.

4.1 Four reasons not to match the DGP

The first constraint is identifiability: forcing a three-component model on a design with insufficient contrasts produces unidentified parameters, not component estimates.

- A pure open-label trial cannot distinguish BR from PB ; both rise from zero together on treatment initiation.
- Fitting three components to a one-contrast design produces implausibly tight standard errors that misrepresent inferential precision.
- The analyst must audit whether the design supplies the necessary contrasts before specifying a component-matched model.

The second constraint is degrees of freedom: a component-matched analysis adds approximately a dozen non-linear parameters, inflating moderation estimand variance at $N \in [30, 150]$.

- Gompertz BR (3), PB ($3 + \text{expectancy multiplier}$), and TV (3) each require participant-specific random effects on the maxima.
- At trial-relevant N , the variance of $\hat{\beta}_{bm:D}$ is dominated by participant and timepoint counts.
- Additional parametric structure typically inflates moderation estimand variance more than it reduces bias.

The third constraint is convergence fragility: component-matched nonlinear mixed models converge below 90% at trial-relevant N , contaminating aggregated analyses.

- Simultaneous Gompertz *BR*, *PB*, and *TV* fitting requires `nlme::nlme`, Bayesian hierarchical, or structural-equation approximations.
- Each is more sensitive to starting values, prone to local optima, and likely to fail on individual trial datasets than the LME analogue.
- Non-converging cells at below-90% convergence rates contaminate power and bias estimates in simulation studies.

The fourth constraint is estimand robustness: the `bm:Dbc` marginal interaction is the right estimand for the biomarker-validation question regardless of the `BR-PB` split.

- Regulators want to know whether the biomarker predicts treatment response under trial conditions, not a mechanistically decomposed prediction.
- The marginal `bm:Dbc` interaction captures this question whether the moderation is biological or expectation-mediated.
- Component decomposition adds mechanistic precision but is not required for the core biomarker-validation decision.

The inferential target is stated as an estimand: the population biomarker-treatment interaction with explicit intercurrent-event handling, while the pharmacological-component target is more ambitious and inherits the heterogeneity-identifiability caveat.

- The $bm:D_{bc}$ estimand is defined by its population, treatment contrast, endpoint, summary, and intercurrent-event strategy in the ICH E9(R1) sense.
- Dropout is handled by a treatment-policy strategy and residual carryover by the modeled exponential decay.
- Recovering a participant-level pharmacological interaction inherits Senn’s caveat that true heterogeneity is separable from within-person variation only under repeated phase contrasts.

4.2 A small extension that usually pays off: phase indicators

The phase-augmented LME uses `Dbc:phase` and `bm:Dbc:phase` to test whether drug and biomarker-by-drug effects attenuate under blinding.

- `phase` absorbs systematic differences in *PB* expectancy across trial phases.
- `Dbc:phase` detects whether the drug effect shrinks under blinding, a signature of *PB*-mediation.
- `bm:Dbc:phase` detects whether the biomarker-treatment interaction shrinks under blinding, a signature that it is partly biomarker-by-*PB*.

The phase-augmented extension adds at most four parameters yet captures the practically relevant belief-attenuation diagnostic without requiring the Gompertz parametric form.

- Three to four parameters is well within budget at any realistic trial sample size.

- The extension does not separately estimate *BR* and *PB* components but identifies the change in apparent drug effect when belief is hedged.
- For most analyses this is the practically relevant question and is answerable without committing to the Gompertz form.

4.3 When to fit the full decomposition

The full decomposition merits the effort only when N is in the hundreds, the design is fully phase-balanced, and the scientific question requires the BR-PB mechanistic split.

- Feasibility conditions: several-hundred N , dense timepoints in every phase, mechanistic BR-PB split required, and informative Bayesian priors available.
- In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or `brms` can return participant-specific component estimates.
- Outside that regime the phase-augmented LME is the better tool.

The DGP-analysis asymmetry is deliberate: simulation uses the rich DGP to characterize how a simpler analysis behaves; the analysis need not mirror the DGP to produce useful inference.

- Simulation controls the DGP and can specify any structure; inference must pay the identifiability cost of every parameter.
- The DGP is a characterization tool; the analysis is a finite-data inference tool; they have different optimality criteria.
- DGP-matching is not required and is often counterproductive at trial-relevant sample sizes.

The exponential-decay carryover indicator is a deliberately parsimonious special case of richer carryover models, and the simulation should benchmark it against a distributed-lag alternative.

- The continuous indicator D_{bc} decays as $\exp(-\lambda t_{sd})$, a one-parameter form tied to a pharmacokinetic half-life.
- Distributed-lag models, the average-period-treatment-effect estimand, and G-estimation relax the single-rate assumption at additional parametric cost.
- The exponential restriction is an assumption, not a finding, and the pre-registered simulation should include a distributed-lag arm to quantify its cost.

4.4 What the one-component analysis estimates

The one-component treatment-effect estimate has a probability limit equal to the pharmacological component plus belief and natural-history contributions, so its bias does not vanish as N grows.

- Writing the reduction as $BR+PB+TV-\varepsilon$ and fitting one drug indicator gives $\text{plim } \hat{\beta}_D = m_{BR} + b_{PB} + b_{TV}$.

- b_{PB} and b_{TV} are the belief and natural-history terms that load onto the drug indicator under the chosen contrast; both are generally non-negative for an improving outcome.
- Because the displacement is in the probability limit, a larger one-component trial converges on the same biased value: the failure is identification, not precision.

The biomarker slope from a one-component analysis decomposes additively into component-specific slopes, so a biomarker correlated with PB or TV registers as a predictor even with no pharmacological association.

- By linearity of covariance, $\beta_{bm}^{\text{lumped}} = \beta_{bm}^{BR} + w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV}$.
- A non-zero β_{bm}^{PB} or β_{bm}^{TV} produces a non-zero one-component slope even when the pharmacological slope β_{bm}^{BR} is exactly zero.
- The result is an algebraic identity under the additive decomposition, not a sample-size-dependent tendency.

The bias displaces the estimand rather than inflating variance, so a significance test of the one-component coefficient cannot detect it; the decomposition is what locates it.

- The one-component coefficient is displaced from m_{BR} by $b_{PB} + b_{TV}$ but remains a well-estimated quantity with a valid Wald test of its own nullity.
- Detecting the bias requires comparing $\hat{\beta}_D$ to m_{BR} , a comparison only the decomposition or an isolating phase contrast supplies.
- This is why a rejection-rate summary is the wrong instrument, and accounts for the flat rejection rates across the placebo-strength axis in the section 7.1 pilot.

4.5 A numerical illustration

The worked example uses two participants with identical one-component responses but opposite BR-PB profiles to show that the one-component analysis cannot distinguish them.

- Participant A has high PB ($m_{PB} = 6$) and moderate BR ($m_{BR} = 4$); Participant B has low PB ($m_{PB} = 1$) and high BR ($m_{BR} = 8$).
- Both participants reach approximately 10–11 points of total improvement by week 8, making them indistinguishable to a one-component analysis.
- The blinded-discontinuation phase creates the diagnostic contrast that exposes the underlying difference.

Open-label totals are clinically indistinguishable; the one-component analysis estimates a population drug effect of approximately 10 points with no differentiation between A and B.

- Both participants show approximately 10–11 points of improvement and present as clinical successes.
- A t-test or simple LME on open-label change cannot detect the heterogeneity in BR versus PB composition.

- The population drug-effect estimate conflates both pharmacological and placebo contributions.

The blinded-discontinuation phase reveals the BR-PB heterogeneity: Participant A retains 4 points while B collapses to 1.5, exposing the diagnostic contrast.

- Participant A's high PB at $\eta = 0.5$ (3 points) sustains improvement even without active drug.
- Participant B's low PB (0.5 points) and absent BR produce near-complete return to baseline.
- The 4.0 versus 1.5 contrast is exactly what the decomposition uses to separate BR from PB .

The one-component model estimates drug at 7.75 points with no principled decomposition, obscuring that A's true BR is 4 and B's is 8.

- The 7.75 point estimate is a conflated average of true BR values (4.0 and 8.0) plus the placebo wash-out.
- The model cannot determine whether A and B differ, nor estimate the pharmacological drug effect separately.
- Placebo and natural-history effects are not identifiable from this contrast alone.

The full decomposition recovers all six component estimates by exploiting phase differential expectancy and the blinded-placebo contrast.

- The LME returns $\hat{m}_{BR}^A = 4.0$, $\hat{m}_{BR}^B = 8.0$, $\hat{m}_{PB}^A = 6.0$, $\hat{m}_{PB}^B = 1.0$ at the population point estimate.
- Participant-level estimates have meaningful uncertainty at trial-relevant N ; the simulation study characterizes this.
- The phase differential ($\eta = 1$ vs. $\eta = 0.5$) is the identification mechanism for the PB parameter.

The one-component model produces one biased average; the three-component model produces six clinically actionable quantities, making biomarker validation possible.

- The one-component estimate of 7.75 points is a biased average of $BR_A = 4$ and $BR_B = 8$ confounded with placebo wash-out.
- The decomposition supplies individual BR , PB , and TV estimates enabling per-participant clinical decisions.
- Predictive-biomarker validation requires the decomposition because biomarker-by- BR is the target estimand.

A population-level example shows that cohort differences in open-label totals driven entirely by PB produce apparent non-replicability that the decomposition resolves.

- Cohort 1 (naive, $m_{PB} = 7$) reports 13-point improvements; Cohort 2 (skeptical, $m_{PB} = 2$) reports 8 points; true $m_{BR} = 5$ in both.
- A one-component analysis would falsely conclude the drug works less well in Cohort 2 and trigger searches for confounders.
- The decomposition returns $\hat{m}_{BR} = 5$ in both cohorts, attributing the difference correctly to PB heterogeneity.

5 Simulation study

The simulation study follows ADEMP and quantifies bias and identifiability properties of three analysis strategies across a prazosin-PTSD-calibrated parameter grid.

- The Morris (2019) ADEMP framework structures the study: Aims, DGP, Estimands, Methods, Performance measures.
- The pre-registration is at `analysis/scripts/component-decomposition/00-ademp-pre-registration`.
- Three analysis strategies (one-component, phase-augmented, full decomposition) are compared across the parameter grid.

5.1 Pilot validation of analysis machinery (100 replicates, 6 cells)

The pilot is a structural check of the analysis pipeline, not a substantive empirical result.

- The realized Study A run (section 7.3) is the empirical core of the manuscript; this pilot precedes it.
- The pilot confirms only that data generation, model fitting, and summary output run without error.
- No bias or power claims are derived from the 100-replicate pilot.

The pilot executed a 6-cell, 100-replicate run with perfect convergence, confirming the pipeline is operational.

- The grid crossed three *pb_strength* levels against two analysis strategies.
- Convergence was 1.0 across all six cells under 8-core `mclapply`.
- Output is checkpointed at `analysis/data/quick-sim/`; no inferential claims are drawn.

Two pilot findings revise the production specification: rank-deficiency forces a formula change, and bias must be read from coefficient means, not rejection rates.

- At $N = 35$, rank-deficiency dropped the D_{bc} :phase term, eliminating the key biomarker-treatment-by-phase contrast.
- The predicted bias manifests in the mean of $\hat{\beta}_{bm:D_{bc}}$, not in the Wald-test rejection rate.
- The production program addresses both by using larger N , a fuller design, and coefficient-mean reporting.

5.2 Production-simulation specification

The production program follows the ADEMP framework with four deliverables; the realized Study A run in section 7.3 is the first of these.

- The deliverables are interaction-estimate bias, full-decomposition identifiability, Type I error and power, and participant-level signal recovery.
- Sample sizes $N \in \{35, 70, 200, 300\}$ cross a *pb_strength* grid; Type I cells target MCSE 0.003 at 5,000 replicates.
- Driver scripts, checkpointed outputs, and the ADEMP pre-registration reside under `analysis/scripts/component-decomposition/`.

5.3 Study A results: bias and power at 1,000 replicates per alternative cell

At 1,000 replicates per alternative cell the one-component estimate of $\hat{\beta}_{bm:D_{bc}}$ is essentially unbiased across the entire (m_{PB}, m_{TV}) grid, contradicting the framework's prediction of a placebo-strength-driven bias.

- The true biomarker-by-*BR* effect is -2.25 ; mean absolute bias is 0.016 at $N = 35$, 0.016 at $N = 70$, 0.006 at $N = 200$, and 0.006 at $N = 300$, in every case within one MCSE of zero.
- Bias shows no monotone trend across $m_{PB} \in \{0, 1, 3, 6, 10\}$ or $m_{TV} \in \{-1, 0, 1, 2\}$; it behaves as sampling noise around zero.
- This is the predicted consequence of the identity in section 6.1: the DGP couples the biomarker to *BR* only, so $\beta_{bm}^{PB} = \beta_{bm}^{TV} = 0$ and the one-component slope is unbiased by construction.

The phase-augmented analysis provides no bias reduction and incurs a substantial power penalty, so it is dominated under this uncontaminated DGP.

- Mean power at $N = 35$ is 0.35 (phase-augmented) versus 0.59 (one-component); at $N = 70$, 0.62 versus 0.90; both reach approximately 1.0 by $N = 200$.
- Phase-augmented mean absolute bias is no smaller than one-component at any N (0.028 versus 0.016 at $N = 35$).
- Convergence is approximately 100% for both analyses at every N (minimum 0.998), so the rank-deficiency that degraded the $N = 35$ pilot does not recur here.

Study A confirms the identity but cannot exhibit the biomarker-validation failure; a contaminated-biomarker pilot reported in the next subsection provides the decisive evidence.

- As pre-registered, Study A varies component magnitude with the biomarker fixed to *BR*, so it probes an axis the identity predicts to be null.
- The decisive axis is $c_{bm,PB}$, the biomarker-*PB* correlation; this is varied in the contaminated-biomarker pilot below.
- On current evidence the phase-augmented analysis is not recommended for the uncontaminated-biomarker regime; the pilot below evaluates it under contamination.

5.4 Contaminated-biomarker pilot (100 replicates per cell)

A pilot varying the biomarker- PB correlation $c_{bm,PB} \in \{0, 0.15, 0.30, 0.45\}$ confirms that bias in the one-component estimator scales with coupling strength, not component magnitude, and that the phase-augmented analysis provides no protection.

- At $N = 300$, mean $\hat{\beta}_{bm:D_{bc}}$ shifts from -2.25 (true) to -1.74 at $c_{bm,PB} = 0.45$ (bias $+0.51$, 30 MCSEs from zero), confirming the identity's prediction.
- The bias is driven by $c_{bm,PB} \times \sigma_{PB}$, not by the population-mean m_{PB} : cells with $m_{PB} = 0$ and $m_{PB} = 6$ show identical bias at every $c_{bm,PB}$ level, because the individual-level PB variance ($\sigma_{PB} = 2$) provides the contamination channel even when the mean is zero.
- The phase-augmented analysis shows the same bias as the one-component analysis at every contamination level, offering no protection against biomarker- PB coupling.
- This pilot uses 100 replicates per cell; a production run at 1,000 replicates per cell is required to confirm the pattern and assess Type I error under contamination.

The contaminated-biomarker pilot reframes the paper's practical guidance: the risk of biomarker-validation failure arises from biomarker- PB correlation in the study population, not from the presence of strong placebo response per se.

- A trial in a population where the biomarker is uncorrelated with placebo responsiveness can use the one-component estimator without bias, regardless of how large the placebo response is.
- A trial where the biomarker covaries with optimism, suggestibility, or treatment-seeking history is at material risk of contamination; bias of $+0.51$ (23% of the true effect) at $c_{bm,PB} = 0.45$ illustrates the practical scale.
- Neither the one-component nor the phase-augmented analysis removes this bias; full component decomposition is required.

5.5 Study B: recovery under a belief-decoupling design

Study B shows the contamination is removable, but only under a design that decouples drug exposure from belief: on a balanced-placebo design a belief-covariate decomposition recovers the unbiased pharmacological slope across the contamination range, while the one-component estimator does not.

- On the standard hybrid design the drug state D_{bc} and the expectancy η are collinear, so no decomposition can separate the channels; we confirmed that several tractable decompositions fail there, no better than the one-component analysis.
- A balanced-placebo design adds a covert on-drug phase (drug given, belief withheld, $\eta = 0$) and an open-placebo phase (drug withheld, belief sustained, $\eta = 1$), which decouples D_{bc} from η .
- The decomposition $Y \sim bm + t + D_{bc} + \eta + bm:D_{bc} + bm:\eta$ then isolates β_{bm}^{BR} on the $bm:D_{bc}$ term while $bm:\eta$ absorbs the placebo channel.

At 1,000 replicates per cell on the balanced-placebo design, the decomposition is essentially unbiased across the contamination range while the one-component estimator degrades monotonically.

- At $N = 300$, the decomposition's bias of $\hat{\beta}_{bm:D_{bc}}$ stays within one MCSE of zero across $c_{bm,PB} \in \{0, 0.1, 0.2, 0.3\}$ ($-0.022, +0.008, +0.021, -0.026$; $MCSE \approx 0.02$), while the one-component bias rises monotonically to $+0.16, +0.34, +0.48$.
- Nominal-95% coverage is near-nominal for the decomposition (0.94–0.96) and falls to 0.86 for the one-component analysis at the highest contamination; convergence is 100 percent for both.
- The contamination grid is capped at $c_{bm,PB} = 0.30$ because beyond approximately 0.45 the implied covariance is non-positive-definite (minimum eigenvalue -0.35 at 0.45) and the contaminated DGP is no longer faithfully representable.

Recovery is therefore conditional on the design, not on the analysis alone: the decomposition is necessary but the belief-decoupling design is what makes the pharmacological slope identifiable.

- The practical implication is that a biomarker-validation trial exposed to possible biomarker- PB coupling should adopt a balanced-placebo or open-hidden design, not the standard hybrid design, if the pharmacological interaction is to be recovered.
- This sharpens the framework's central claim: decomposition is the mechanism, but a design that varies belief independently of drug exposure is the enabling condition.
- Open questions remaining: recovery beyond the positive-definite range, the participant-level component recovery of the original Study B identifiability aims, and replication of the pattern under the mean-moderation architecture (mean-moderation) coupling.

6 Application to predictive-biomarker validation

The precision-medicine question – whether a biomarker predicts pharmacological response – is answered by the one-component total only when the biomarker is uncorrelated with PB and TV ; otherwise it requires decomposing the response.

- The question is whether the biomarker predicts the BR component, not the one-component total.
- The PATH statement formalizes this as the target estimand for predictive biomarker validation.
- The BR-PB-TV decomposition is the mechanism that separates the pharmacological signal when biomarker-component coupling cannot be ruled out.

A one-component-total biomarker regression conflates pharmacological and placebo-belief prediction, producing a stratification biomarker that mis-identifies responders.

- $\hat{\beta}_{bm}$ from the one-component regression contains both pharmacological and placebo-belief prediction in unknown proportions.

- If PB covaries with the biomarker distribution, flagged 'high responders' include high- PB responders who will not benefit pharmacologically.
- Prospective application of such a biomarker would systematically mis-stratify patients for prescription decisions.

A valid pharmacological predictor requires a non-zero $\hat{\beta}_{bm}^{BR}$; significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ indicate a psychological or prognostic biomarker, not a pharmacological one.

- The sole criterion for pharmacological prediction is a meaningfully non-zero BR -component regression coefficient.
- A biomarker correlating only with PB selects placebo responders, not drug responders.
- A biomarker correlating only with TV is prognostic but not predictive in the pharmacological sense.

When biomarker-component coupling cannot be ruled out, the decomposition is the mechanism by which the precision-medicine question is posed; when it can, the one-component analysis already answers it.

- Where the biomarker may correlate with PB or TV , separating BR from them is what prevents validation from conflating pharmacological and non-pharmacological prediction.
- The phase-augmented `bm:Dbc:phase` interaction is a small- N proxy, but Study A shows it buys nothing under an uncontaminated biomarker, so it is warranted only when contamination is plausible.
- The full decomposition recovers component-specific regressions explicitly when sample size permits.

7 Discussion

The three-component framework reframes treatment response as a structured sum of causally distinct contributions, and the present work maps when acting on that decomposition actually changes inference.

- The framework specifies which trial-design contrasts identify which components.
- Acting on the decomposition is valuable conditionally: for attribution estimands, and for biomarkers that may correlate with PB or TV , but not for a biomarker-treatment interaction with a biomarker orthogonal to them.
- The target estimand for biomarker validation is the biomarker-by- BR association; whether a one-component analysis already recovers it depends on the biomarker's coupling to PB and TV .

Four open questions bound the framework's current scope: the simulation results are pending, the η multipliers require sensitivity analysis, non-monotone natural history is not yet handled, and participant-level recovery inherits the identifiability caveats of individual treatment effects.

- Quantitative bias-versus-power trade-offs are conjectures until the section 7 simulation is run.
- The η (phase) expectancy-modulation parameters are modeling assumptions requiring explicit sensitivity analysis.
- Cyclical or biphasic TV trajectories exceed the Gompertz form and require flexible or covariate-augmented specifications.
- Participant-level component recovery rests on separating participant-by-component signal from participant-by-period and participant-by-phase variation, the identifiability caveat Senn raises for individual treatment effects.

Two companion documents extend the manuscript: a pedagogical narrative and a Gompertz-family sensitivity evaluation.

- The pedagogy document at `docs/24-component-decomposition-pedagogy.md` provides worked examples for first-time readers.
- The Gompertz evaluation at `analysis/report/07-gompertz-evaluation/` quantifies trajectory-specification sensitivity.
- Together they address the intuitive and parametric dimensions of the framework that the present manuscript treats briefly.

The components exist in any longitudinal treatment response, but participant-level identifiability requires a within-subject belief-by-exposure contrast that not every placebo-controlled design supplies.

- The three-component structure is a property of the DGP of any repeated-measures treatment study; its identifiability is design-specific.
- Parallel-group placebo control identifies the population-mean PB but not a participant-level PB , and supplies no within-subject belief contrast; only designs with such a contrast (blinded or open-hidden discontinuation, randomized withdrawal, crossover, hybrid N-of-1) identify the participant-level decomposition the biomarker question needs.
- The broad claim is therefore scoped to designs supplying a within-subject belief-by-exposure contrast, not to placebo-controlled longitudinal trials in general.

8 Conclusions

Treatment response in aggregated N-of-1 trials is the sum of three causally distinct components, and the bias from conflating them is governed by the estimand and the biomarker's coupling to the components, not by their raw magnitude.

- The three components are BR (pharmacological), PB (placebo-belief), and TV (natural history).
- Each has its own biological mechanism, identifiability conditions, and clinical interpretation.
- The one-component pre-post treatment effect is biased by the PB and TV contributions; the biomarker-treatment slope is biased only by the biomarker's covariance with PB and

876 TV , as Study A confirms.

877 **What matters is the design contrast, not the parametric model: the trial must**
878 **supply the drug-on/off and belief contrasts, but adding parametric component**
879 **terms is not free and is not always worth it.**

- 880 • Unique component identification requires phase contrasts built into the trial design; the
881 one-component analysis already exploits the drug-on/off contrast.
- 882 • Under an uncontaminated biomarker, the phase-augmented analysis adds no bias reduc-
883 tion and loses power, so it is not recommended as a default.
- 884 • The phase-augmented and full component-matched analyses are warranted for attribu-
885 tion estimands and for biomarkers that may correlate with PB or TV , where the extra
886 structure does work.

887 **For predictive-biomarker validation the decomposition matters precisely when the**
888 **candidate biomarker may correlate with PB or TV ; the contaminated-biomarker**
889 **study demonstrates this failure and Study B shows it is recoverable under a**
890 **belief-decoupling design.**

- 891 • A biomarker that predicts the one-component total is a pharmacological predictor only if
892 it is uncorrelated with PB and TV ; otherwise the one-component slope is contaminated.
- 893 • Regulatory and prescribing decisions therefore require separating BR from PB and TV
894 whenever biomarker-component coupling cannot be ruled out.
- 895 • Study A established the uncontaminated baseline, the contaminated-biomarker study
896 (section 7.4) demonstrated the failure, and Study B (section 7.5) showed the bias is
897 recoverable on a balanced-placebo design but not on the standard hybrid design.

898 9 Reproducibility

899 **The manuscript is rendered inside a Docker container with locked package ver-**
900 **sions; the full software environment is pinned by Dockerfile and renv.lock.**

- 901 • R 4.4.x in the zzcollab Docker container provides the execution environment.
- 902 • Simulation packages are `nlme`, `parallel`, `data.table`, `furrr`, and `purrr`.
- 903 • The manuscript build uses `rmarkdown` and `bookdown`.

904 **Reproducibility is ensured by a master seed of 20260507 with per-replicate seeds**
905 **derived by index addition.**

- 906 • `set.seed(20260507)` is called at the top of each replicate loop.
- 907 • `parallel::mc.set.seed` propagates seeds inside `mclapply`.
- 908 • Per-replicate seeds are `master_seed + rep_idx`.

909 **All simulation outputs, driver scripts, pre-registration, and manuscript source**
910 **files are versioned at documented paths within the repository.**

- 911 • Per-replicate outputs and Morris summaries are at `analysis/data/quick-sim/`.

- 912 • Driver scripts and the ADEMP pre-registration are at `analysis/scripts/component-decomposition/`.
- 913 • Manuscript source, bibliography, and CSL file are at `analysis/report/06-component-decomposition/`.